

# TFT DISPLAY SPECIFICATION



RAYSTAR

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## RFU700D4-AYH-MNG1

### General Specifications

- Size: 7.0 inch
- Dot Matrix: 800× RGB ×1280 dots
- Module dimension: 131.63 x 188.17 x 8.05 mm
- Active area: 94.2 x 150.72 mm
- Pixel pitch: 0.11775 x 0.11775 mm
- LCD type: TFT, Normally Black, Transmissive
- Viewing angle: 85/85/85/85
- TFT Driver IC: ILI9881C or Equivalent
- TFT Interface: 2-Lanes MIPI
- Aspect Ratio: 16:9
- Backlight Type: LED ,Normally White
- PCAP Driver IC: ILI2511
- PCAP Interface: I2C
- PCAP FW Version: 0x06.0x00.0x00.0x00.0x77.0x88.0x00.0x02
- PCAP Resolution: 16384\*16384
- Touch Panel: Projected Capacitive Touch Panel, PCAP
- Surface: Glare

# Interface

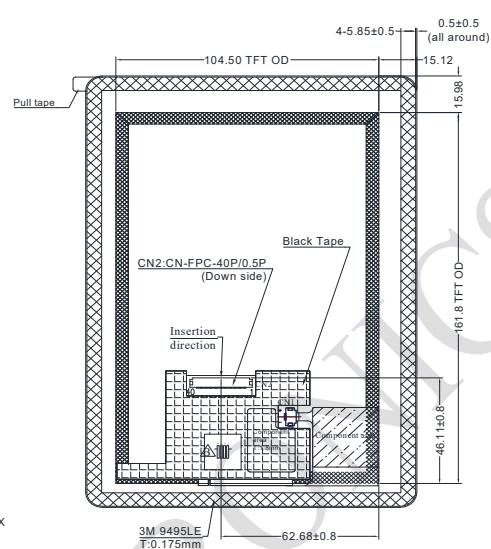
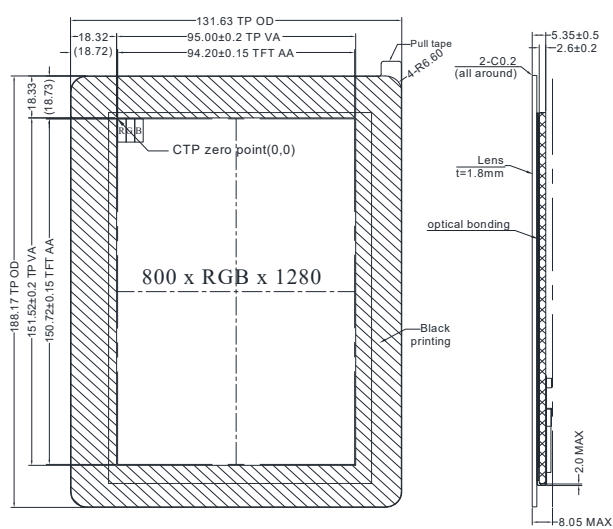
## LCM PIN Definition

| Pin   | Symbol | Function                                                                               |
|-------|--------|----------------------------------------------------------------------------------------|
| 1     | TP_GND | Ground                                                                                 |
| 2     | TP_SDA | TP serial interface address and data input/output.                                     |
| 3     | TP_SCL | TP serial interface clock input.                                                       |
| 4     | TP_INT | TP indicate coordinate data ready.                                                     |
| 5     | TP_RST | System reset signal input, active low.                                                 |
| 6     | NC(YU) | No connection                                                                          |
| 7     | NC(XL) | No connection                                                                          |
| 8     | NC(YD) | No connection                                                                          |
| 9     | NC(XR) | No connection                                                                          |
| 10-11 | VCI    | Power for analog circuit.                                                              |
| 12-13 | NC     | No connection                                                                          |
| 14    | RESET  | Global reset pin.                                                                      |
| 15    | TE     | Tearing effect output pin.<br>Leave the pin open when not in use.                      |
| 16    | NC     | No connection                                                                          |
| 17-18 | GND    | Ground                                                                                 |
| 19-20 | IOVCC  | Power supply for I/O system.                                                           |
| 21    | GND    | Ground                                                                                 |
| 22-23 | NC     | No connection                                                                          |
| 24    | GND    | Ground                                                                                 |
| 25-26 | NC     | No connection                                                                          |
| 27    | GND    | Ground                                                                                 |
| 28    | CLKP   | MIPI DSI differential clock pair leave it open or fix to LVDSVSS level when not in use |
| 29    | CLKN   | MIPI DSI differential clock pair leave it open or fix to LVDSVSS level when not in use |
| 30    | GND    | Ground                                                                                 |
| 31    | D1P    | MIPI DSI differential data pair.                                                       |
| 32    | D1N    | MIPI DSI differential data pair.                                                       |

|       |       |                                  |
|-------|-------|----------------------------------|
| 33    | GND   | Ground                           |
| 34    | D0P   | MIPI DSI differential data pair. |
| 35    | D0N   | MIPI DSI differential data pair. |
| 36-37 | GND   | Ground                           |
| 38    | VLED+ | LED input terminal (Anode)       |
| 39-40 | VLED- | Ground (Cathode)                 |

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# Contour Drawing



| CN2     |          |
|---------|----------|
| PIN NO. | PIN Name |
| 1       | TP_GND   |
| 2       | TP_SDA   |
| 3       | TP_SCL   |
| 4       | TP_INT   |
| 5       | TP_RST   |
| 6       | NC(YU)   |
| 7       | NC(XL)   |
| 8       | NC(YD)   |
| 9       | NC(XR)   |
| 10      | VCI      |
| 11      | VCI      |
| 12      | NC       |
| 13      | NC       |
| 14      | RESET    |
| 15      | TE       |
| 16      | NC       |
| 17      | GND      |
| 18      | GND      |
| 19      | IOVCC    |
| 20      | IOVCC    |
| 21      | GND      |
| 22      | NC       |
| 23      | NC       |
| 24      | NC       |
| 25      | NC       |
| 26      | NC       |
| 27      | GND      |
| 28      | CLKP     |
| 29      | CLKN     |
| 30      | GND      |
| 31      | D1P      |
| 32      | D1N      |
| 33      | GND      |
| 34      | D1P      |
| 35      | D1N      |
| 36      | GND      |
| 37      | GND      |
| 38      | VED+     |
| 39      | VED-     |
| 40      | VED+     |

The non-specified tolerance of dimension is  $\pm 0.3$  mm .

### Absolute Maximum Ratings

| Item                  | Symbol | Min | Typ | Max | Unit |
|-----------------------|--------|-----|-----|-----|------|
| Operating Temperature | TOP    | -20 | —   | +70 | °C   |
| Storage Temperature   | TST    | -30 | —   | +80 | °C   |

### Electrical Characteristics

#### Typical Operation Conditions

| Item               | Symbol | Values |      |      | Unit | Remark      |
|--------------------|--------|--------|------|------|------|-------------|
|                    |        | Min.   | Typ. | Max. |      |             |
| Supply voltage     | VCI    | 2.5    | 3.3  | 3.6  | V    |             |
|                    | IOVCC  | 1.7    | 1.8  | 1.9  | V    |             |
| Supply LCM current | Ivci   | -      | 130  | 195  | mA   | Include CTP |
|                    | IioVCC | -      | 18   | 30   | mA   |             |

#### Backlight Driving Conditions

| Parameter     | Symbol | Min. | Typ.   | Max. | Unit |
|---------------|--------|------|--------|------|------|
| LED current   | ILED   | -    | 140    | -    | mA   |
| LED voltage   | VLED+  | 16.8 | 18.0   | 19.2 | V    |
| LED Life Time |        | -    | 30,000 | -    | Hr   |